# Multilevel Spatial Feature-Based Manifold Metric Learning for Domain Adaptation in Remote Sensing Image Classification

Yanni Dong, *Senior Member, IEEE*, Xuexiang Qin, Xue Li, and Lina Xu

***Abstract*— Domain adaptation purposes to transfer well-labeled domain knowledge to poor-labeled domain. However, many domain adaptation methods focus too much on spectral features and simple spatial features, causing failure to eliminate noise that is an important risk in remote sensing image classification. Degenerated features are also a potential risk to feature transformation in domain adaptation. In addition, it is also a typical problem that reduces the difference in probability distribution between different domains for domain adaptation. To alleviate these issues, we propose a multilevel spatial feature-based manifold metric learning (MSF-MML) method for domain adaptation in remote sensing image classification. First, it applies and iterates spatial information filtering to extract multilevel spatial features (MSFs) by calculating the average of sample pixels and neighboring pixels, effectively obtaining more stable feature information. Then, it utilizes the Grassmann manifold space to alleviate the problem of degenerated features of remote sensing data in domain adaptation. Finally, it exploits maximum mean discrepancy (MMD) to construct metric constraints that reduce discrepancy in domain probability distributions. The effectiveness of MSF-MML was demonstrated on three real-world datasets.**

***Index Terms*— Domain adaptation, Grassmann manifold, metric learning (ML), multilevel spatial features (MSFs).**

## I. Introduction

REMOTE sensing image classification plays an important role in land cover classification, lithological geological exploration, environmental monitoring, and other fields [1], [2], [3], [4], [5], [6]. With the advancement of sensor technology, the data volume of remote sensing images has increased dramatically, while it is quite scarce with well-labeled samples, that is, manually labeling remote sensing images requires a huge cost, so it is particularly important to obtain labeled samples inexpensively and efficiently [7], [8]. For remote sensing, the influence of factors, such as light angles, sensor differences, atmospheric variations, and seasonal changes, may lead to large differences between images acquired in different geographical areas or at different times in the same block [9]. Therefore, if the existing classifiers are directly applied to new datasets, they usually fail to get satisfactory classification results [10]. Domain adaptation technique is considered to solve this problem. The initial data with well-labeled information are called the source domain, while the data with poor or no labeled information are called the target domain. Domain adaptation assumes that the source and target domains have the same task, but the distribution of domains exists visibly distinct [11], [12], [13]. In the context of this assumption, domain adaptation attempts to align the distribution between the domains and applies the classifier learned from the source domain to the target domain, thus achieving labeling of the target domain [14], [15], [16], [17], [18]. Domain adaptation has received a lot of attention from scholars, who have modeled many approaches from different perspectives.

Based on a consensus, domain adaptation approaches have been classified into three categories: 1) instance-based methods; 2) model-based methods; and 3) feature-based methods [19], [20]. We will focus on introducing feature-based methods as they are the main focus of our study. We further classify feature-based domain adaptation methods into three categories: 1) statistical feature-based methods; 2) geometric feature-based methods; and 3) joint statistical and geometric feature methods.

The statistical feature-based methods consider alignment statistics, such as mean and variance, to align the distribution of features between domains. In statistical feature-based methods, the maximum mean discrepancy (MMD) is the most widely used distance metric. Huang et al. [21] constructed kernel mean matching (KMM) and Pan et al. [22] constructed transfer component analysis (TCA) that they both used MMD and kernel functions to match marginal distributions in the source and target domains. Long et al. [23] expanded TCA by making use of transfer joint matching (TJM), which not only reduced the difference between the source and target domains' marginal distributions but also measured the importance of the source domain samples. Since TCA only addressed the problem of marginal distribution, Wang et al. [24] and Long et al. [25] extended the problem to address both marginal

Manuscript received 3 January 2023; revised 5 July 2023 and 9 August 2023; accepted 22 September 2023. Date of publication 13 October 2023; date of current version 23 October 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62222116 and Grant 62171417; in part by the Project Supported by the Open Fund of Hubei Luojia Laboratory under Grant 220100058; and in part by the Key Laboratory of Intelligent Computing and Signal Processing, Ministry of Education, Anhui University, under Grant 2020B002. *(Yanni Dong and Xuexiang Qin are co-first authors.) (Corresponding author: Lina Xu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3323579

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distribution and conditional distribution by balanced distribution adaptation (BDA) and joint distribution adaptation (JDA), respectively. Sun et al. [26] presented a "frustratingly easy" unsupervised domain adaptation method called correlation alignment (CORAL). The CORAL aligned the input feature distributions of the source and target domains by exploring their second-order statistics. In the context of remote sensing image classification, Wang et al. [27] introduced a technique in which each class was assigned a weight in the process of minimizing the intraclass distance to mitigate the impact of class imbalance. Wang et al. [28] jointly embedded domain-related alignments (CORAL) and class-related alignments to achieve the JDA performance.

The geometric feature-based methods consider the possible spatial geometric structure of the aligned data. From the viewpoint of MMD, Fernando et al. [29] attempted to find a variation matrix to reduce the difference between the subspaces of source and target domains to achieve the goal of subspace alignment (SA), which constructed the subspaces of domains by principal component analysis (PCA). Sun and Saenko [30] developed the SA to minimize the distribution difference of the subspaces while reducing the difference between the subspaces of source and target domains. The Grassmann manifolds are also a useful technique for domain adaptation. Gopalan et al. [31] considered subspaces of domains (of the same dimension) as points in Grassmann manifolds and used sampling geodesic flow (SGF) to obtain a description of the domain transfer. Gong et al. [32] wanted to apply the geodesic flow kernel (GFK) to overcome the problem of quantity selection of SGF subspace. The GFK regarded the domains as points in manifold space, and the process of transfer was treated as the integration of the geodesic between the points.

The joint statistical and geometric feature methods consider the simultaneous alignment of both statistical and geometric features between the domains. Wang et al. [33] introduced a method called manifold embedded distribution alignment (MEDA), which extended adaptation regularization based transfer learning (ARTL) [34] into Grassmann manifold space based on GFK and integrated the dynamic distribution alignment (DDA), the structural risk minimization (SRM), and the Laplacian regularization (LR). Zhang et al. [35] developed a method that utilizes two coupled projections to reduce both the geometrical shift and distribution shift simultaneously, by projecting the data from the source domain and target domain into low-dimensional subspaces. In the context of remote sensing image classification, Huang et al. [36] introduced a method called graph embedding and distribution alignment (GEDA), which joined distribution alignment and SA techniques to minimize the spectral differences between domains. Dong et al. [37] proposed the spatial–spectral weighted kernel manifold embedding distribution alignment (SSWK-MEDA), which applied a novel spatial information filtering and then constructed a composite kernel with dynamic weights of spatial kernel and spectral kernel.

All the above methods show excellent classification performance, but there are still some major challenges that should not be ignored as follows.

1) First, limited by imaging conditions, such as illumination conditions and data transmission errors, it frequently occurs that the original spectral features have noise [38], [39], [40], [41], which poses challenges for the classifier's work. Most methods focus on the use of spectral information or simple spatial features of remote sensing data, but they are not sufficient to smooth out the effects of noise.
2) Furthermore, in the context of domain adaptation, many feature transformation-based methods revolve around the concept of changing the projection, according to [33] and [42]. Consequently, a situation arises where certain samples exhibit excellent classification performance in the original feature space. However, when subjected to projection transformation, their features may undergo changes, resulting in a degradation of classification performance.
3) Finally, it has been a classic problem that aligns the distribution between the domains for domain adaptation [64], but the previous methods may have an inadequate ability to tackle the problem of distribution differences, which decreases the classification performance of the classifier.

To alleviate these problems, we propose a method called multilevel spatial feature-based manifold metric learning (MSF-MML) for domain adaptation in remote sensing image classification. First, MSF-MML defines a filter window, where sample pixels in the same block of remote sensing images usually have the same labels and similar spectral features, to calculate the average spectrum of sample pixels and iterate the filter window for extracting multilevel spatial features (MSFs). These MSFs represent these class properties well and can be used to directly replace the original spectral features, which can effectively smooth out the classification burden added by noise. Then, MSFs are mapped into Grassmannian manifolds with the help of GFK, a kind of classical subspace learning method based on Grassmannian manifolds, and are alleviated from the negative effects of degenerated features in the original feature space by the specific geometrical structure of Grassmannian space. Finally, MSF-MML constructs metric constraints on intraclass and interclass distances. Because there is an absently suitable method that can quantitatively assess the importance of intraclass and interclass distances, MSF-MML innovatively minimizes the interclass divergence with the help of MMD to correct the metric constraints, which better aligns the distribution of domains.

The main contributions are summarized as follows.

1) A defined filter window is iterated to obtain MSFs for smoothing the negative effects of noise in the original spectral features.
2) The MSFs are mapped into the Grassmannian manifold, which alleviates the degenerated features of remote sensing data in domain adaptation.
3) The metric constraints are constructed on the interclass and intraclass distances, which innovatively minimize the interclass divergence and reduce distribution differences between domains.

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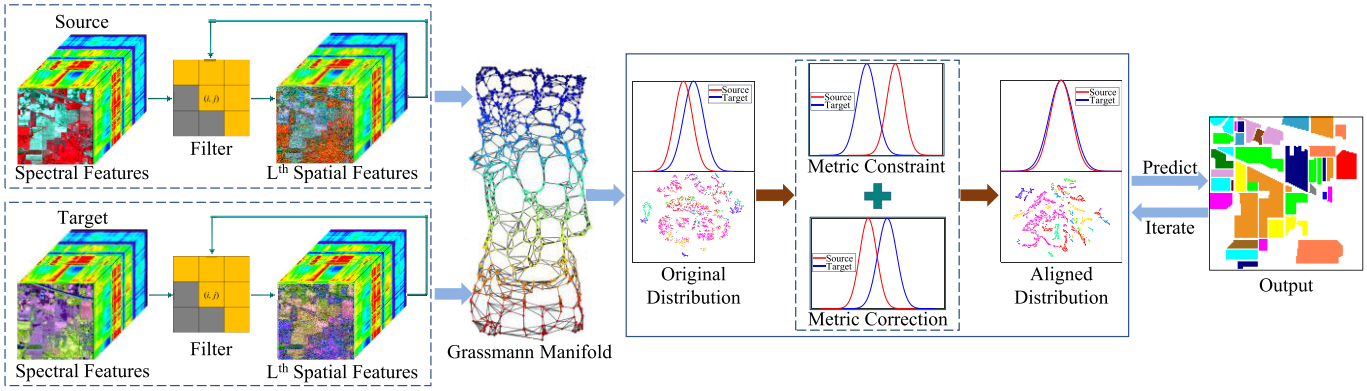



Fig. 1. Flowchart of the proposed MSF-MML.

The remainder of this article is organized as follows. First, related works are presented in Section II. Then, the details of the proposed method are described in Section III. Next, the experimental designs and experimental results as well as the analysis are presented in Section IV. Finally, conclusions and directions of future works are outlined in Section V.

## II. Related Work

### A. Multilevel Spatial Features

Ma et al. [43] applied a filter window to extract the average spectrum of the sample pixels, and the intention was to replace the original spectrum with a stable spectral. Huang et al. [36] also agreed that remote sensing images can be spatially correlated. Therefore, they directly used spatial filtering as a pre-processing step for remote sensing images, which was used to maintain similarity between neighboring pixels and to maintain consistency in the local neighborhoods. Similar to these approaches, we also think that spatial features of remote sensing images should receive more attention. Thus, MSF-MML extracts the average spectrum of the sample pixels by filter window, and more reliable spectral features are obtained by iterating the filter window, which is called MSFs.

### B. Degenerated Features

If the domains are only transformed by features or subspace aligned, the problem of degenerated features cannot be eliminated. Sometimes, a low-dimensional latent subspace may remove or reduce the effects of degenerated features in the original feature space [42]. The features in the Grassmann manifold, which has some geometrical structures [44], [45], can alleviate degenerated features in the original space. MSF-MML uses the GFK as an effective trick to alleviate the problem of degenerated features.

### C. Metric Learning

For domain adaptation, it is a classical issue that the distribution of domains exists visibly distinct. Li et al. [46] jointly utilized feature adaptation for distribution matching and sample adaptation with landmark selection. They also considered local consistency between samples in the knowledge transfer. Huang and Zhou [47] utilized MMD to construct unsupervised metric transfer learning (UTML), which constrained the intraclass and interclass distances of the domains. MSF-MML also uses MMD to construct metric constraints on intraclass and interclass distances inspired by UTML, which reduces probability distribution differences of the domains. Moreover, MSF-MML innovatively minimizes interclass divergence to improve the robustness of the classifier.

## III. Methodology

### A. Problem Definition

Given a labeled source domain image $D_s = \{\mathbf{X}_s^{\text{spe}}, \mathbf{Y}_s\}$ and an unlabeled target domain image $D_t = \{\mathbf{X}_t^{\text{spe}}\}$, where subscripts $s$ and $t$ indicate the source domain and the target domain, respectively, $\mathbf{X}_{s/t}^{\text{spe}} \in \mathbb{R}^{H_{s/t} \times W_{s/t} \times \dim}$ is the size and spectral dimension of the image, where $H_{s/t}$ and $W_{s/t}$ represent the source/target image sizes and dim denotes the spectrum dimension. The source domain and target domain contain $n_s$ and $n_t$ samples. In addition, $C$ represents the number of classes. The two domains are assumed to be homogeneous, that is, the feature space $\mathbf{X}_{s/t}^{\text{spe}} \in \mathbb{R}^{n_s/n_t \times \dim}$ and the label space $\mathbf{Y}_{s/t} \in \mathbb{R}^C$. Our goal is to find a classifier $f$, which can predict the labels of the target domain by using knowledge from the source domain. The flowchart of the MSF-MML is shown in Fig. 1.

### B. Spatial Information

For remote sensing images, there is inevitable noise in the original spectral information limited by the imaging conditions, which greatly reduces the representativeness of the spectral features for each class. However, it is a natural advantage that the same class in geospatial space is usually grouped into clusters that are incomparable to other data. Meanwhile, the class features are highly similar in clusters. In remote sensing images, it is common for not all pixels to be labeled. To enhance the efficiency of MSF-MML, only the labeled pixels are utilized in extracting spatial features. Therefore, a filter window of $3 \times 3$ size is applied to all labeled sample pixels to obtain spatial features. In Fig. 2,

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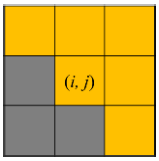



Fig. 2. Example of filter window. The orange cells represent labeled pixels, and the gray cells represent unlabeled pixels.

we show an example filter window where each cell represents one pixel. The orange cells represent labeled pixels, and the gray cells represent unlabeled pixels. In this article, only the orange cells in the filter window are required to participate in the subsequent work.

The central cell $(i, j)$ that is required to be labeled samples the current position of the filter window. The spatial features $\mathbf{X}_{i,j}^{\text{spa}} \in \mathbb{R}^{1\times \text{dim}}$ of the central cell are equal to the average of the spectral features $\mathbf{X}_{p,q}^{\text{spe}} \in \mathbb{R}^{1\times \text{dim}}$ of all orange cells. MSF-MML refers to the spatial features extracted at this time as first-level spatial features. For each sample pixel $\mathbf{X}_{p,q}^{\text{spe}}$, the filter window has the following equation in each band:

$$\mathbf{X}_{i,j}^{\text{spa}} = \frac{1}{h}\sum_{p,q} \mathbf{X}_{p,q}^{\text{spe}}, \quad y_{p,q} \neq 0 \tag{1}$$

where $h$ is the number of sample pixels in the filter window, $y_{p,q}$ is the label of pixel $\mathbf{X}_{p,q}^{\text{spe}}$, and $p$ and $q$ are the positions relative to the central cell.

When extracting the spatial features of the target domain, MSF-MML only needs to know whether it belongs to sample pixels and does not need to exactly know which precise class each sample belongs to. Therefore, it does not matter whether the target domain has a precise class label at this stage.

MSF-MML applies the same filter window for the first-level spatial features to obtain the second-level spatial features. With the same idea, MSF-MML iterates this process based on the second-level spatial features to obtain the $L$-level spatial features (where $L$ is a hyperparameter). The work is shown in Fig. 3. In this article, MSF-MML will directly refer to the $L$-level spatial features as MSFs $\mathbf{X}^{\text{spa}}$ for convenience. These MSFs can effectively smooth out the presence of noise, exhibiting more representativeness compared to the original spectral features. The original spectral feature $\mathbf{X}^{\text{spe}}$ is transformed into MSFs $\mathbf{X}^{\text{spa}}$ by filtering.

### C. Manifold Embedded

The Grassmann manifold space has been shown to have some special geometrical structures that can effectively alleviate the problem of degenerated features in the original space [44], [45]. GFK is an effective trick to map MSFs into Grassmann manifold space. It has good classification performance and lower time costs, which help improve the accuracy of soft labels and save time. Therefore, GFK is combined with our work to alleviate the situation of degenerate features. Only the main idea of GFK is introduced here, and more details can be found in the original paper [32].

In the GFK, first, the PCA subspaces $\mathbf{P}_s$ and $\mathbf{P}_t \in \mathbb{R}^{l\times d}$ of the source and target domains were regarded as two different points $\Phi(0)$ and $\Phi(1)$ in the Grassmann manifold. Then, the integral method was used to find a geodesic line $\{\Phi(t) : 0 \le t \le 1\}$ from $\Phi(0)$ to $\Phi(1)$ and to calculate the GFK $\mathbf{G}$

$$\langle z_i, z_j \rangle = \int_0^1 (\Phi(t)^T \mathbf{x}_i)^T (\Phi(t)^T \mathbf{x}_j)^T dt = \mathbf{x}_i^T \mathbf{G} \mathbf{x}_j. \tag{2}$$

Finally, the features of the original space were transformed into the Grassmann manifold by $z = g(\mathbf{x}) = \sqrt{\mathbf{G}}\mathbf{x}$.

In this article, the MSFs are applied in (2) to construct the GFK $\mathbf{G}^{\text{spa}}$. Then, MSF-MML uses $\mathbf{Z}^{\text{spa}} = \sqrt{\mathbf{G}^{\text{spa}}}\mathbf{X}^{\text{spa}}$ to map $\mathbf{X}_s^{\text{spa}}$ and $\mathbf{X}_t^{\text{spa}}$ into manifold space and obtain the corresponding manifold MSFs $\mathbf{Z}_s^{\text{spa}}$ and $\mathbf{Z}_t^{\text{spa}}$ in the source and target domains. MSF-MML directly replaces the original MSFs $\mathbf{X}_s^{\text{spa}}$ and $\mathbf{X}_t^{\text{spa}}$ with the manifold MSFs $\mathbf{Z}_s^{\text{spa}}$ and $\mathbf{Z}_t^{\text{spa}}$ to complete the subsequent task.

### D. Metric Learning

For domain adaptation, the goal is that well-labeled source domain knowledge is transferred to the target domain lacking labels, so it is necessary to have a more accurate and robust classifier. Naturally, it is important to increase the distances between different classes and decrease the distance between the same class. However, the label of the target domain is required to be sufficient, which is contradictory to the reality that there is no label with the target domain. In fact, many works have already begun to address this issue [48], [49], [50], [51]. Here, MSF-MML utilizes the soft labels of the target domain, which is obtained by using an easy classifier applied from the source domain to the target domain. The easy classifier is required to have good accuracy and lower time costs, such as K-Nearest Neighbor (KNN) and support vector machine (SVM).

Another key point of domain adaptation is what kind of appropriate method to describe the distance of samples or the distance between domains. There are already many methods to answer this question, such as MMD [52], [53] and Kullback–Leibler (KL) divergence [54], [55], [56]. In this article, MMD is chosen to measure the distance of samples and the distance between domains. On the one hand, MMD is a nonparametric metric, which effectively avoids the annoyance caused by parametric problems. On the other hand, it is a sample mean metric that is not sensitive to label noise, which satisfies the current situation where only soft labels are available in the target domain. In addition, traditional metrics are very sensitive to noisy data from soft labels [47]. The MMD distance between distributions $p$ and $q$ is defined as

$$\text{Dist}^2(p, q) = \left\| E_p[\psi(x)] - E_q[\psi(x)] \right\|_{\mathbf{H_K}}^2 \tag{3}$$

where $\mathbf{H_K}$ is the reproducing kernel Hilbert space (RKHS) induced by the feature map $\psi(\cdot)$. Here, $E[\cdot]$ denotes the mean of the embedded samples.

In order to construct an MMD associated with classifier $f$, MSF-MML adopts projected MMD to calculate the intraclass distance and interclass distance. Then, the process can be

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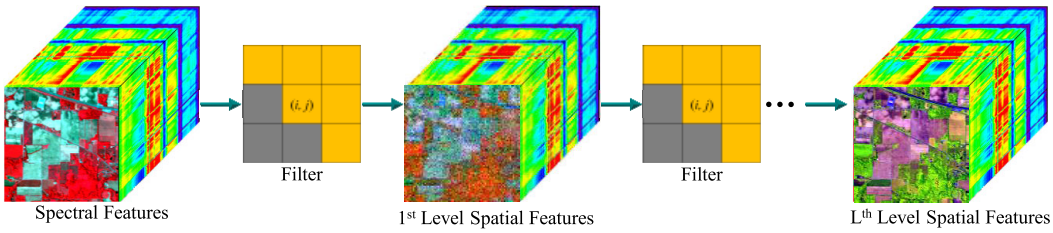



Fig. 3. Extracting MSFs.

expressed as

$$\begin{aligned}\mathbf{M}_f\left(\mathbf{Z}_s^{\text{spa}}, \mathbf{Z}_t^{\text{spa}}\right) &= \sum_{c=1}^{C}\left\|E\left[f\left(\left(\mathbf{z}_s^{\text{spa}}\right)^{(c)}\right)\right]-E\left[f\left(\left(\mathbf{z}_t^{\text{spa}}\right)^{(c)}\right)\right]\right\|_{\mathbf{H_K}}^2 \\ &\quad - \sum_{c=1}^{C}\left\|E\left[f\left(\left(\mathbf{z}^{\text{spa}}\right)^{(c)}\right)\right]-E\left[f\left(\left(\mathbf{z}^{\text{spa}}\right)^{(\tilde{c})}\right)\right]\right\|_{\mathbf{H_K}}^2 \quad (4)\end{aligned}$$

where $(\mathbf{z}_s^{\text{spa}})^{(c)}$ and $(\mathbf{z}_t^{\text{spa}})^{(c)}$ are the samples of class $c$ in the source and target domains, respectively, $(\mathbf{z}^{\text{spa}})^{(c)}$ is the sample of class $c$ in all samples, and $(\mathbf{z}^{\text{spa}})^{(\tilde{c})}$ does not belong to class $c$ in all samples.

To balance the possible overconstraint of the metric constraints, MSF-MML uses projection MMD to minimize the interclass divergence. Then, the process can be expressed as

$$\mathbf{H}_f\left(\mathbf{Z}_s^{\text{spa}}, \mathbf{Z}_t^{\text{spa}}\right) = \frac{1}{C}\sum_{c=1}^{C}\left\|E[f((\mathbf{z}^{\text{spa}})^{(c)})]-E[f(\mathbf{z}^{\text{spa}})]\right\|_{\mathbf{H_K}}^2 \quad (5)$$

where $(\mathbf{z}^{\text{spa}})^{(c)}$ is the sample of class $c$ in all samples and $\mathbf{z}^{\text{spa}}$ is the full sample.

The relative importance of constructing intraclass distances and interclass distances is not equal for different tasks. Therefore, if the MMD is solely utilized to increase the distances between different classes and decrease the distances between the same classes, the robustness of the classifier cannot be guaranteed. However, we unable to find a suitable metric that precisely weighs the significance of these distances. As a result, we assume that in the absence of a suitable metric to measure intraclass and interclass distances, metric constraints are likely to lead to overconstraining of metric learning (ML). To address this issue, we introduce a metric correction term that minimizes the interclass divergence to alleviate the overconstraint and achieve a balanced ML process. The purpose of this correction term is to compensate for the limitation of lacking a precise measure of the importance of intraclass and interclass distances, thereby improving the robustness and generalization of the classifier. Finally, the metric correction in MSF-MML is expressed as minimizing the interclass divergence, which is based on our experimental findings. It is undeniable that there may be other ways to express the metric correction, which still requires more experiments to discover.

### E. Classifier Construction

After manifold feature (MF) learning and ML, the classifier $f$ can be learned by summarizing SRM over $D_s$, metric constraint, and metric correction. The classifier $f$ can be represented as

$$\begin{aligned}f = \arg\min_{f\in\mathbf{H_K}} \sum_{i=1}^{n_s}\left(y_i - f\left(\mathbf{z}_i^{\text{spa}}\right)\right)^2 + \eta\|f\|_{\mathbf{K}}^2 \\ + \lambda\mathbf{M}_f\left(\mathbf{Z}_s^{\text{spa}}, \mathbf{Z}_t^{\text{spa}}\right) + \rho\mathbf{H}_f\left(\mathbf{Z}_s^{\text{spa}}, \mathbf{Z}_t^{\text{spa}}\right). \quad (6)\end{aligned}$$

In order to understand MSF-MML more effectively, the details will be shown about (6) in the following.

*1) Structural Risk Minimization:* The representer theorem [57] is able to optimize the classifier $f$

$$f(\mathbf{z}^{\text{spa}}) = \sum_{i=1}^{n_s+n_t}\boldsymbol{\beta}_i\mathbf{K}\left(\mathbf{z}_i^{\text{spa}}, \mathbf{z}^{\text{spa}}\right) \quad (7)$$

where $\boldsymbol{\beta} = (\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \ldots)^T \in \mathbb{R}^{(n_s+n_t)\times C}$ is the coefficient of the constructed kernel element $\mathbf{K}$. Then, (6) can be rewritten as follows:

$$\begin{aligned}f = \arg\min_{f\in\mathbf{H_K}}\left\|(\mathbf{Y}-\boldsymbol{\beta}^T\mathbf{K})\mathbf{A}\right\|_F^2 + \eta\text{tr}(\boldsymbol{\beta}^T\mathbf{K}\boldsymbol{\beta}) \\ + \lambda\text{tr}(\boldsymbol{\beta}^T\mathbf{K}\mathbf{M}\mathbf{K}\boldsymbol{\beta}) + \rho\text{tr}(\boldsymbol{\beta}^T\mathbf{K}\mathbf{H}\mathbf{K}\boldsymbol{\beta}). \quad (8)\end{aligned}$$

In (8), the first two terms represent SRM for minimizing the classification error, where $\|\cdot\|_F$ is the Frobenius norm. $\mathbf{K} \in \mathbb{R}^{(n_s+n_t)\times(n_s+n_t)}$ is the kernel matric with $\mathbf{K}_{ij} = \mathbf{K}(\mathbf{z}_i^{\text{spa}}, \mathbf{z}_j^{\text{spa}})$, and $\mathbf{A} \in \mathbb{R}^{(n_s+n_t)\times(n_s+n_t)}$ is a diagonal label indicator matrix with each element $\mathbf{A}_{ii} = 1$ if $\mathbf{z}_i^{\text{spa}} \in D_s$, and $\mathbf{A}_{ii} = 0$ otherwise. $\mathbf{Y} = [y_1, \ldots, y_{n_s+n_t}] \in \mathbb{R}^{(n_s+n_t)\times C}$ is the label matrix. It is no matter that the target labels are unknown since they are filtered out by the label indicator matrix $\mathbf{A}$. $\text{tr}(\cdot)$ denotes the matrix trace operation.

*2) Metric Constraint:* In (8), the third term is the metric constraint term, where $\mathbf{M}$ is a merged matrix with intraclass distance $\sum_{c=1}^{C}\mathbf{M}_s^{(c)}$ and interclass distance $\sum_{c=1}^{C}\mathbf{M}_d^{(c)}$

$$\mathbf{M} = \sum_{c=1}^{C}\mathbf{M}_s^{(c)} - \sum_{c=1}^{C}\mathbf{M}_d^{(c)}. \quad (9)$$

The elements of $\mathbf{M}_s^{(c)}$ and $\mathbf{M}_d^{(c)}$ are given as follows:

$$\left(\mathbf{M}_s^{(c)}\right)_{ij} = \begin{cases}\dfrac{1}{\left(n_s^{(c)}\right)^2}, & z_i, z_j \in \left(\mathbf{Z}_s^{\text{spa}}\right)^{(c)} \\ \dfrac{1}{\left(n_t^{(c)}\right)^2}, & z_i, z_j \in \left(\mathbf{Z}_t^{\text{spa}}\right)^{(c)} \\ -\dfrac{1}{n_s^{(c)}n_t^{(c)}}, & \begin{cases} z_i \in \left(\mathbf{Z}_s^{\text{spa}}\right)^{(c)}, & z_j \in \left(\mathbf{Z}_t^{\text{spa}}\right)^{(c)} \\ z_i \in \left(\mathbf{Z}_t^{\text{spa}}\right)^{(c)}, & z_j \in \left(\mathbf{Z}_s^{\text{spa}}\right)^{(c)} \end{cases} \\ 0, & \text{otherwise}\end{cases} \quad (10)$$

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$$\left(\mathbf{M}_d^{(c)}\right)_{ij} = \begin{cases} \dfrac{1}{(n^{(c)})^2}, & z_i^{\text{spa}}, z_j^{\text{spa}} \in (\mathbf{Z}^{\text{spa}})^{(c)} \\ \dfrac{1}{(n^{(\tilde{c})})^2}, & z_i^{\text{spa}}, z_j^{\text{spa}} \notin (\mathbf{Z}^{\text{spa}})^{(c)} \\ -\dfrac{1}{n^{(c)} n^{(\tilde{c})}}, & \begin{cases} z_i^{\text{spa}} \in (\mathbf{Z}^{\text{spa}})^{(c)}, & z_j^{\text{spa}} \notin (\mathbf{Z}^{\text{spa}})^{(c)} \\ z_i^{\text{spa}} \notin (\mathbf{Z}^{\text{spa}})^{(c)}, & z_j^{\text{spa}} \in (\mathbf{Z}^{\text{spa}})^{(c)} \end{cases} \end{cases} \tag{11}$$

where $n_s^{(c)}$ and $n_t^{(c)}$ are the sample numbers of class $c$ in the source and target domains, respectively. $n^{(c)} = n_s^{(c)} + n_t^{(c)}$ and $n^{(\tilde{c})} = n_s + n_t - n^{(c)}$. We train an easy classifier to obtain soft labels that alleviate the lack of labels in the target domain.

*3) Metric Correction:* In (8), the final term is the metric correction term, where $\mathbf{H}$ aims to correct the metric constraint of (9) by minimizing the interclass divergence

$$\mathbf{H} = \frac{1}{C}\sum_{c=1}^{C}\left(\frac{1}{n^c}\mathbf{1}_{n^c}\mathbf{1}_{n^c}^T - \frac{1}{n}\mathbf{1}\mathbf{1}^T\right) \tag{12}$$

where $\mathbf{H}$ is the central matrix of intraclass divergence of class $c$. $\mathbf{1}_{n^c} \in \mathbb{R}^{n\times 1}$ is an indicator vector, indicating which samples belong to the current class, that is, if the $i$th sample belongs to class $c$, $\mathbf{1}_{n^c}(i)$ is 1, and otherwise, it is 0, and $\mathbf{1} \in \mathbb{R}^{n\times 1}$ is a vector whose elements all are 1.

By calculating the partial derivative of $\boldsymbol{\beta}$ for (8) and setting it to 0, we can obtain the solution of $\boldsymbol{\beta}$. Then, the final solution can be summarized as follows:

$$f = \mathbf{K}\left(\left((\mathbf{A} + \lambda\mathbf{M} + \rho\mathbf{H})\mathbf{K} + \eta\mathbf{I}\right)^{-1}\mathbf{A}\mathbf{Y}^T\right) \tag{13}$$

where $\mathbf{I} \in \mathbb{R}^{(n_s+n_t)\times(n_s+n_t)}$ is an identity matrix and $f$ is an $(n_s + n_t) \times C$ matrix. Each row represents a sample which each of its columns represents one of the classes. Elements in the matrix represent the possibility of the corresponding sample belonging to the class. We consider the highest possibility class of each sample as its label, which achieves the predicted labels of the target domain. The MSF-MML algorithm is described in Algorithm 1.

**Algorithm 1** Procedure of the MSF-MML Method

**Input:** Source domain image $D_s = \{\mathbf{X}_s^{spe}, \mathbf{Y}_s\}$, target domain image $D_t = \{\mathbf{X}_t^{spe}\}$, manifold subspace dimension $d$, regularization coefficient $\lambda, \eta, \rho$, repeat times $T$, $L^{th}$ level spatial features $L = 5$.

**Output:** classifier $f$.

**Steps:**

1. $\mathbf{X}_s^{spa} = \mathbf{X}_s^{spe}$ and $\mathbf{X}_t^{spa} = \mathbf{X}_t^{spe}$.
2. **Repeat**
3. Apply the spatial filter window Eq.(1) to the source domain $\mathbf{X}_s^{spa}$ and target domain $\mathbf{X}_t^{spa}$. Update multilevel spatial features $\mathbf{X}_s^{spa}$ and $\mathbf{X}_t^{spa}$.
4. **Until $\mathbf{L}^{\text{th}}$ level**
5. Map multilevel spatial features $\mathbf{X}_s^{spa}$ and $\mathbf{X}_t^{spa}$ into manifold space using Eq.(2), and obtain manifold spectral features $\mathbf{Z}_s^{spa}$ and $\mathbf{Z}_t^{spa}$.
6. Train a simple classifier on $\mathbf{Z}_s^{spa}$ to obtain the soft labels of $\mathbf{Z}_t^{spa}$.
7. **Repeat**
8. Calculate the coefficient matrix of metric constraint $\mathbf{M}$ by using Eq.(9).
9. Calculate the coefficient matrix of metric correction $\mathbf{H}$ by using Eq.(12).
10. Learn classifier $f$ by using Eq.(13).
11. Update the soft labels of $\mathbf{Z}_t^{spa}$.
12. **Until $T$ times.**
13. Return predict labels $\mathbf{Y}_t$ of target domain.

## IV. Experiments

### A. Data Description

The first dataset is the hyperspectral data of grss_dfc_2013, also called the Houston dataset [58]. It was acquired by the National Centre for Airborne Laser Mapping (NCALM) using the ITRES CASI-1500 hyperspectral imager. The Houston-2013 data were collected in 2012 and released by the IEEE GRSS Data Fusion Contest (DFC) in 2013. The hyperspectral imagery has 144 spectral bands in the 380-1050nm region with 349 × 1905 in size and a spatial resolution of 2.5 m, which is acquired over the University of Houston campus and the neighboring urban area. Based on the difference in illumination conditions, the image is divided into two parts, named Bright and Shadow. According to our attempts, in order to achieve a balance between computational efficiency and visualization of qualitative results, we have chosen to focus on 8 out of the total 15 classes. These classes include healthy grass, stressed grass, residential, trees, commercial, road, highway, and railway. The bright part has 8541 samples, and the shadow part has 1434 samples. The image and ground truth (GT) of the Houston dataset are shown in Fig. 4.

The second dataset is the hyperspectral data of the Indian Pines data, also called the Indian Pines dataset [59]. It was gathered by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor of the National Aeronautics and Space Administration. The hyperspectral imagery has 200 spectral bands in the 375-2200nm region with pixels and a spatial resolution of 20 m, which is acquired over the Indian Pines test site in Northwestern Indiana. According to the geographical layout of the classes, the image is divided into two parts, named big and small. There are 16 classes in this dataset, and only seven classes are selected to fulfill our work. These classes include Corn-notill, Corn-mintill, Corn, Grass-pasture, Sybeans-clean, Bldg-Grass-Tree-Drives, and Soybeans-mintill. The big part has 4966 samples, and the small part has 1446 samples. The image and GT of the Indian Pines dataset are shown in Fig. 5.

The third dataset is the multispectral data of the Worldview-2 satellite, also called the Worldview-2 dataset [60], [61]. The multispectral imagery has eight spectral bands (blue, green, yellow, red, coastal, near infrared, near infrared, and red edge) with 200 × 200 pixels and a spatial resolution of 1.8 m, which in July 2011 and July 2012 over the same area in Wuhan, China. Based on the difference in time conditions, the image is divided into two parts, named 2011 and 2012. The dataset has a total of four classes, which include red building, forest, gray building, and white building. All four classes of

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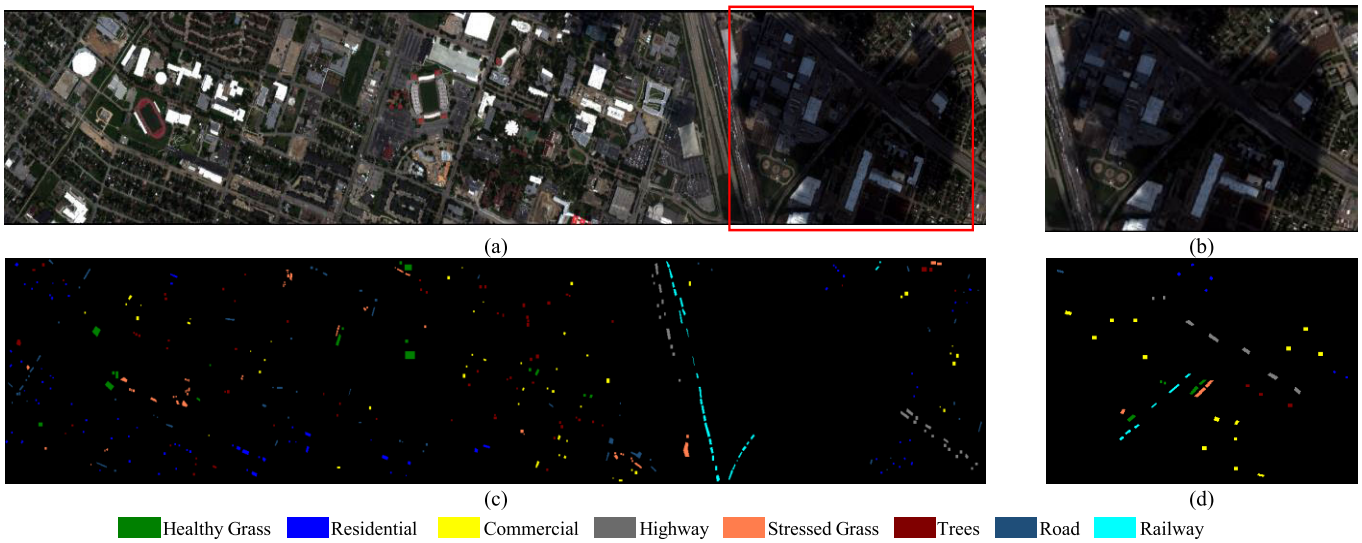



Fig. 4. Image (R:59, G:40, and B:23) and GT of the Houston dataset. (a) Image of the bright part. (b) Image of the shadow part. (c) GT of the bright part. (d) GT of the shadow part.

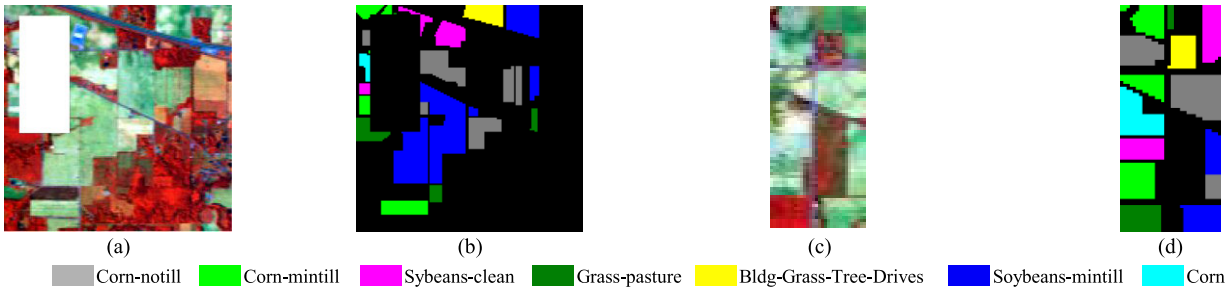



Fig. 5. Image (R:80, G:116, and B:10) and GT of the Indian Pines dataset. (a) Image of the big part. (b) GT of the big part. (c) Image of the small part. (d) GT of the small part.

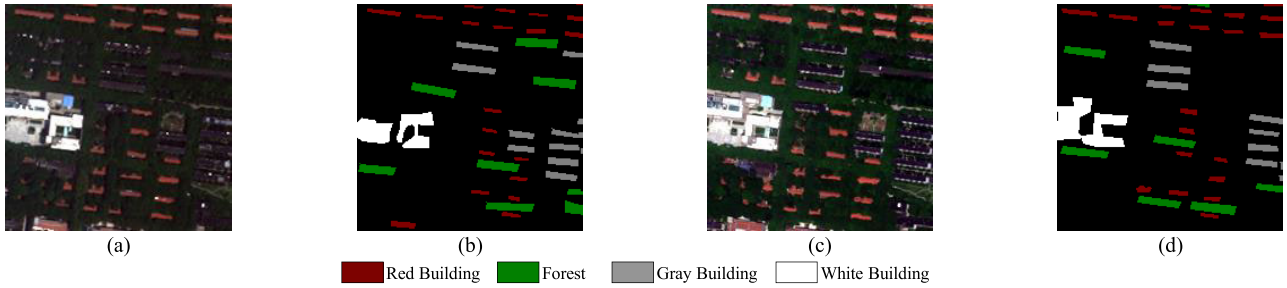



Fig. 6. Image (R:5, G:3, and B:2) and GT of the subset of the Worldview-2 dataset. (a) Image of the time 2011. (b) GT of the time 2011. (c) Image of the time 2012. (d) GT of the time 2012.

samples are required to participate in the following work. The 2011 part has 5317 samples, and the 2012 part has 5425 samples. The image and GT of the Worldview-2 dataset are shown in Fig. 6.

### B. Experimental Setting

*1) Evaluation Indicators:* The main evaluation indicators used in this experiment are overall accuracy (OA) and kappa coefficient (kappa), which are most used in remote sensing image classification.

The OA is the number of correctly classified samples $N_{\text{right}}$ as a percentage of the total number of samples $N$. It is obvious that the OA ranges from 0% to 100%, and the higher the value, the better the classification performance. OA can be expressed as

$$\text{OA} = \frac{N_{\text{right}}}{N} \times 100\%. \tag{14}$$

The numerator of the kappa coefficient is calculated by multiplying the total number of samples $N$ by the number of correctly classified samples $N_{\text{right}}$ and subtracting the sum of the products of the number of samples in each class $N^c$ and the number of samples assigned to that class $N^c_{\text{pre}}$. The denominator of the kappa coefficient is calculated by squaring the total number of samples $N$, subtracting the sum of the

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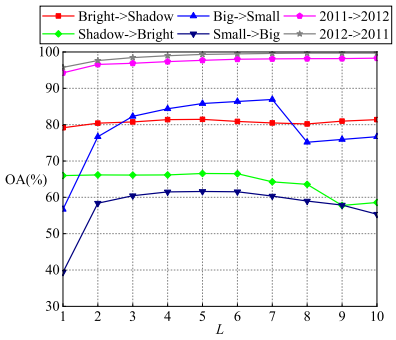



Fig. 7. OAs of MSF-MML under different $L$ values.

product of the number of samples in each class $N^c$, and the number of samples assigned to that class $N^c_{\text{pre}}$. It ranges from $-1$ to 1, with a value closer to 1 indicating better classification performance

$$\text{kappa} = \frac{\text{NN}_{\text{right}} - \sum_{c=1}^{C} N^c N^c_{\text{pre}}}{N^2 - \sum_{c=1}^{C} N^c N^c_{\text{pre}}}. \tag{15}$$

In this article, one-nearest neighbor (1NN) is chosen as an easy classifier in all experiments due to its advantages of simple understanding of how it works, good classification performance, and lower time costs. The 1NN works that the distance between the sample to be classified and all known samples needs to be calculated and sorted, which is assigned to the known class with the closest distance.

*2) Kernel Parameters:* When constructing the kernel function, the radial basis function (RBF) kernel is chosen as the kernel function, and its bandwidth is determined by the variance of the input. The bandwidth of the RBF kernel for the manifold multilevel space features is determined by the variance of the manifold multilevel space features. Thus, the bandwidth of the RBF kernel does not need to be tuned as a hyperparameter.

*3) Levels Parameters:* The level $L$ of MSFs is a very important parameter. Thus, the trend of OA with $L$ from 1 to 10 is shown in Fig. 7. For the Houston dataset, the OA first is a slight fluctuation by $L$. However, when $L$ is large, there is a significant decline. For the Indian Pines dataset, the OA is greatly affected by $L$ in which these lines show a change of rising first and then falling. For the Worldview-2 dataset, there is first a significant rise in OA, but after $L = 3$, this change decreases and begins to converge. This is able to find that OA has good performance when $L = 5$ in all tasks. Finally, $L = 5$ is established as a level of MSFs.

We would like to verify the influence of $L$, so some scatter plots are used to approximately describe the distribution of samples in high-dimensional space with $L$ changes. Taking the Worldview-2 dataset as an example, Fig. 8 shows the scatter plots of the 0–5 level MSFs. These points are reduced from high-dimensional to 2-D by t-SNE [62], [63]. Each point represents a sample, where the same color belongs to the same class and the different colors belong to different classes. While the level $L$ of MSFs increases, we can clearly observe that the same class is becoming more and more compact in the same block, and meanwhile, the distance between different classes is increasingly distant. Therefore, these samples of different classes are becoming increasingly distinguishable.

*4) Dimension Parameters:* When we use GFK to project MSFs into Grassmann manifold space, PCA is usually employed to create subspaces of source and target by reduction dimension. Therefore, dimension parameters $d$ also have an important influence on the classification performance of MSF-MML. Due to the computational requirements of GFK, these dimensional parameters can be set to at most half of the original dimensionality of the MSFs. For the Houston dataset and the Indian Pines dataset, these dimensional parameters are set with parameters ranging from 5 to 70 in steps of 5. However, since the Worldview-2 dataset has only eight bands, its dimension parameter can only be set in the range of 1–4 with a step of 1. Finally, the effect of the dimension parameters on the OA is shown in Fig. 9.

In Fig. 9(a) and (b), for the Houston dataset, the dimension parameters first show a slight increase and then show a stable situation. Therefore, the dimension parameters $d$ corresponding to two tasks of the Houston dataset are 65 and 10. For the Indian Pines dataset, the result of OA is obviously affected by the dimension parameters. In the "Big→Small" task, this line is clearly visible that the overall trend is increasing and then decreasing, with small local fluctuations. However, in the "Small→Big" task, the line change is approximately the opposite as in the "Big→Small" task. The line clearly shows that the general trend is down, with small local fluctuations. Thus, the dimension parameters $d$ corresponding to two tasks of the Indian Pines dataset are 40 and 5, respectively. Dimension parameters $d$ corresponding to the Houston data are 65 and 10. In Fig. 9(b), for the Worldview-2 dataset, the changes in the lines of these two tasks are the easiest to analyze that they all show that the rate of increase at $d = 2$ has an obvious change and a clear overall upward trend. Finally, the dimension parameters $d$ corresponding to two tasks of the Worldview-2 dataset are 4.

*5) Iteration Parameters:* The negative effect of soft labels can be effectively eliminated by iteration. Therefore, the influence of iteration parameters $T$ on OA is shown in Fig. 10(a). The trend can be observed that these lines displayed a slight rise and then a steady state in most tasks. However, the two tasks in the Houston dataset are slightly different in that they first show a small fluctuation and then show a steady state. Observe all six tasks, and it is very easy to notice that the initial stabilization point common to all tasks is position $T = 5$. Finally, $T = 5$ is set as the iteration parameter for all tasks. On the one hand, it shows excellent classification performance. On the other hand, it can maximize significant time cost savings.

*6) Regularization Parameters:* The main regularization parameters of MSF-MML are $\lambda$ for SRM, $\eta$ for metric constraint and $\rho$ for metric correction, whose values are set in the ranges 0.0001, 0.001, 0.01, 0.1, 1, 2, 4, and 8.

In Fig. 10(b), it is presented that the lines change between the value of OA and the parameters $\lambda$ of SRM. We regret that obvious regularities have not been found about these lines. It is almost impossible to find a common parameter that satisfies all tasks, so the best performing parameters are

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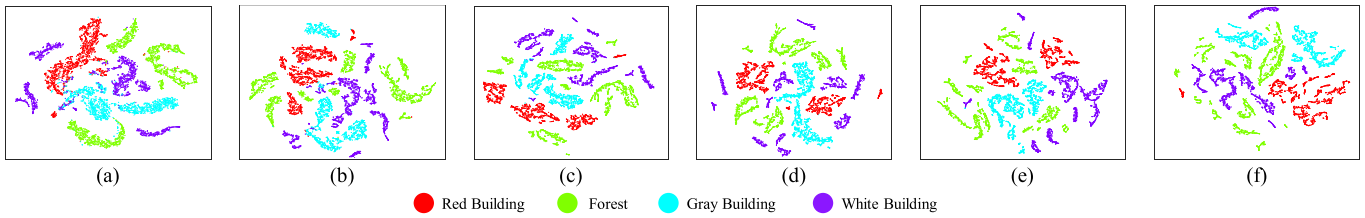



Fig. 8. Scatter plots of MSF-MML for different values of $L$. (a) Spectral (equal to $L = 0$). (b) First level. (c) Second level. (d) Third level. (e) Fourth level. (f) Fifth level.

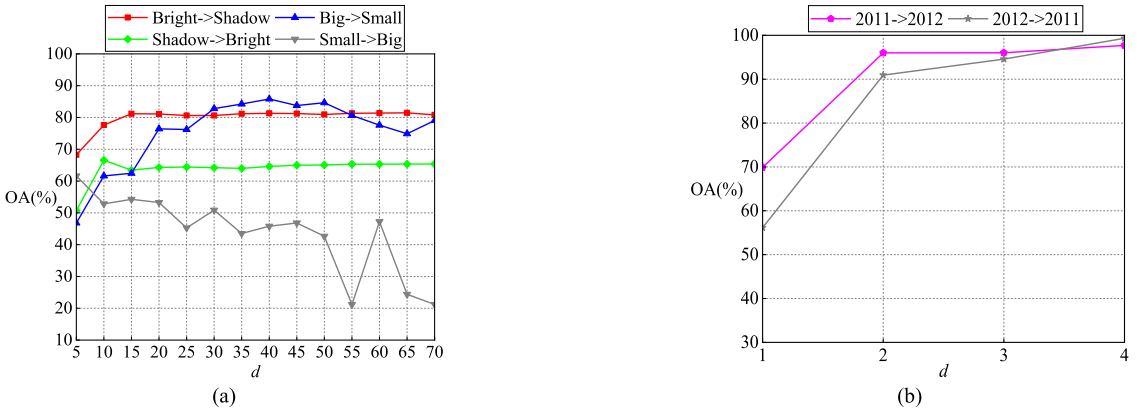



Fig. 9. OAs of MSF-MML under different values of $d$. (a) Houston and Indian Pines datasets. (b) Worldview-2 dataset.

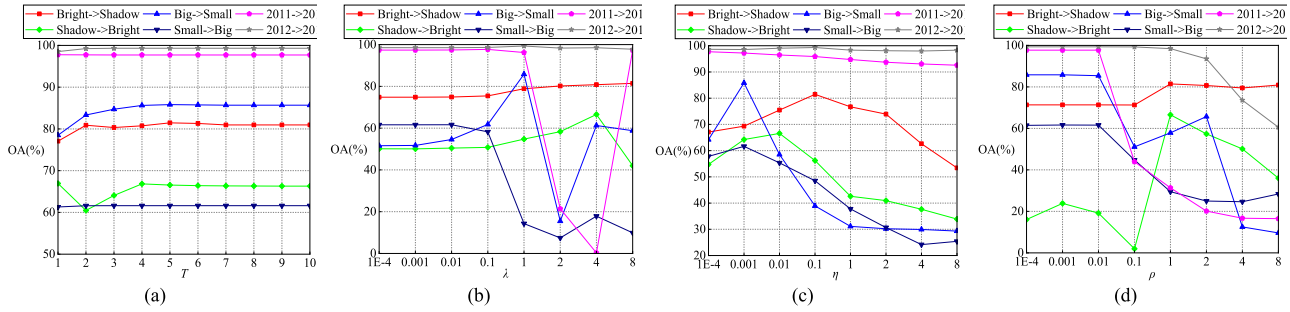



Fig. 10. OAs of MSF-MML under different values of (a) $T$, (b) $\lambda$, (c) $\eta$, and (d) $\rho$.

selected to train the classifier. Finally, the parameters $\lambda$ of SRM corresponding to two tasks of the Houston dataset are 8 and 4. The corresponding parameters $\lambda$ for the two tasks in the Indian pine dataset are 1 and 0.01. The parameters $\lambda$ are set to 0.1 and 1 in the two tasks of the Worldview-2 dataset.

In Fig. 10(c), it is shown that the lines vary between the value of OA and the parameters $\eta$ of metric constraint. In the Houston dataset, the lines are the trend of change that shows an upward and then downward trend. Therefore, the parameters $\eta$ of metric constraint corresponding to two tasks of the Houston dataset are 0.1 and 0.01. In the Indian Pines dataset, the trend of these lines is similar to the Houston dataset, also showing an upward and then a downward trend. When the parameter $\eta$ of metric constraint is 0.001, both tasks of the Indian pine dataset show good classification performance. In the Worldview-2 dataset, their lines show a steady state and then a slight decline. When the parameters $\eta$ of the metric constraint are kept small, two tasks have superior classification performance in the Worldview-2 dataset. Thus, the parameters $\eta$ of the metric constraint corresponding to two tasks of the Worldview-2 dataset are 0.001 and 0.1.

In Fig. 10(d), it is displayed that the lines vary between the value of OA and parameters $\rho$ of metric correction. In the Houston dataset, the trends of the lines are quite different, but both tasks exhibit good classification ability for a value of $\rho$ at 1. In the Indian Pines dataset, these lines show a tendency to plateau and then decline, so if the value of $\rho$ is kept small, there will be good classification results. Thus, the parameters $\rho$ of metric correction corresponding to two tasks of the Indian Pines dataset are 0.0001 and 0.001. In the Worldview-2 dataset, the trend of these lines is similar to the Indian Pines dataset, also showing a steady and then a downward trend. Similarly, when the value of $\rho$ is kept small, there will be good classification performance. Thus, while the value of $\rho$ is 0.0001, both tasks of the Worldview dataset show superior outcomes.

*7) Ablation Experiments:* MSF-MML includes three key components, namely, MSF, MFs, and ML with metric

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TABLE I
OA(%) CLASSIFICATION ACCURACIES OF EACH COMPONENT OF MSF-MML

| Dataset | Task | SF+MF+ML | MSF+ML | MSF+MF | MSF+MF+ML | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | **M+H** | **M+H** | **1NN** | $\mathbf{M}_s$ | $-\mathbf{M}_d$ | **M** | $(1-\mu)\mathbf{M}_s-\mu\mathbf{M}_d$ | **M-H** | **M+H** |
| Houston | Bright->Shadow | **82.08** | 82.01 | 38.84 | 65.61 | 64.93 | 68.95 | 71.51 | 79.01 | 81.45 |
| | Shadow->Bright | 62.51 | 65.41 | 31.16 | 36.31 | 40.56 | 40.19 | 37.02 | 61.40 | **66.56** |
| Indian Pines | Big->Small | 59.82 | 78.49 | 51.38 | 82.85 | 76.14 | 85.82 | **86.93** | 86.51 | 85.82 |
| | Small->Big | 50.77 | 59.32 | 51.65 | 61.46 | 61.32 | 61.46 | **62.44** | 61.84 | 61.62 |
| Worldview-2 | 2011->2012 | 87.67 | **99.96** | 97.53 | 97.42 | 97.53 | 98.19 | 98.06 | 98.34 | 97.71 |
| | 2012->2011 | 88.49 | 99.57 | 86.18 | 99.32 | 98.76 | 99.32 | **99.77** | 99.36 | 99.32 |
| Average | | 71.89 | 80.79 | 59.46 | 73.83 | 73.21 | 75.66 | 75.96 | 81.08 | **82.08** |
| Variance | | 218.02 | 237.41 | 585.94 | 485.99 | 421.47 | 445.90 | 482.43 | 237.22 | **202.56** |

TABLE II
OA(%) CLASSIFICATION ACCURACIES WITH 1NN CLASSIFIER ON DIFFERENT DATASETS

| Dataset | Task | JDA | UTML | GFK | MEDA | GDEA | SSWK-MEDA | MSF-MML |
|---|---|---|---|---|---|---|---|---|
| Houston | Bright->Shadow | 61.72 | 67.43 | 37.31 | 75.87 | 62.83 | 79.85 | **81.45** |
| | Shadow->Bright | 50.01 | 56.41 | 21.53 | 51.74 | 61.35 | 61.07 | **66.56** |
| Indian Pines | Big->Small | 52.97 | 52.63 | 25.10 | 51.11 | 60.72 | 64.45 | **85.82** |
| | Small->Big | 32.18 | 38.02 | 17.26 | 42.07 | 51.95 | 58.98 | **61.62** |
| Worldview-2 | 2011->2012 | 91.10 | 84.87 | 62.38 | 87.28 | 96.85 | 94.69 | **97.71** |
| | 2012->2011 | 80.91 | 73.26 | 50.12 | 86.36 | 95.52 | 96.18 | **99.32** |

TABLE III
KAPPA CLASSIFICATION ACCURACIES WITH 1NN CLASSIFIER ON DIFFERENT DATASETS

| Dataset | Task | JDA | UTML | GFK | MEDA | GEDA | SSWK-MEDA | MSF-MML |
|---|---|---|---|---|---|---|---|---|
| Houston | Bright->Shadow | 0.5530 | 0.6144 | 0.2103 | 0.7150 | 0.5699 | 0.7583 | **0.7767** |
| | Shadow->Bright | 0.4292 | 0.5011 | 0.1014 | 0.4521 | 0.5556 | 0.5550 | **0.6179** |
| Indian Pines | Big->Small | 0.4368 | 0.4335 | 0.1411 | 0.4103 | 0.5326 | 0.5754 | **0.8297** |
| | Small->Big | 0.1989 | 0.2464 | 0.1039 | 0.3040 | 0.3492 | 0.4597 | **0.4840** |
| Worldview-2 | 2011->2012 | 0.8809 | 0.7967 | 0.5006 | 0.8292 | 0.9587 | 0.9289 | **0.9694** |
| | 2012->2011 | 0.7438 | 0.6405 | 0.3612 | 0.8153 | 0.9389 | 0.9482 | **0.9908** |

constraints **M** and metric correction **H**. For convenience, the spectral features are replaced by SF. Under the same parameter range, we conduct experiments to analyze the performance of each component of MSF-MML, and the optimal classification results for each task are presented in Table I. In Table I, some symbols being used need to be clearly explained: **M** represents a joint metric constraint that minimizes intraclass distance $\mathbf{M}_s$ and maximizes intraclass distance $\mathbf{M}_d$. **H** represents the defined metric correction, which minimizes interclass divergence. Finally, we would like to try to manually tune the importance of interclass distance and intraclass distance, so $\mu$ is set as a weight from 0 to 1 in steps of 0.1. In particular, we use the positive sign to express minimization and the negative sign to express maximization in Table I. In Table I, it is evident that MSF-MML only achieves the highest classification results for the "Shadow->Bright" task when compared to the combined performance of the other components. However, it does obtain the highest average value of 82.08 and the smallest variance of 202.56. By considering the average and variance, it becomes evident that the generalization ability of MSF-MML surpasses that of other components.

### C. Performance

In this section, six tasks are conducted to verify the effectiveness of the proposed MSF-MML. Several classic and state-of-the-art algorithms have been employed for comparison algorithms, including JDA [25], UTML [47], GFK [32], MEDA [33], GEDA [36], and SSWK-MEDA [37]. First, we have considered classical domain adaptation methods such as JDA, UTML, and GFK as our comparison methods. These methods serve as a benchmark to evaluate the performance of our proposed approach. Second, we also include domain adaptation methods such as UTML, GFK, and MEDA as comparative methods because they are relevant to our work and have demonstrated superior classification performance. Third, SSWK-MEDA and GEDA are chosen as comparative methods since they are considered to be newer approaches. Quantitative evaluation, qualitative evaluation, and time complexity are employed to evaluate the classification performance. The OA, kappa, and seconds are used as evaluation indicators.

*1) Quantitative Evaluation:* In Tables II and III, we show the OA and kappa performance of MSF-MML and the comparison methods on the Houston dataset, the Indian Pines

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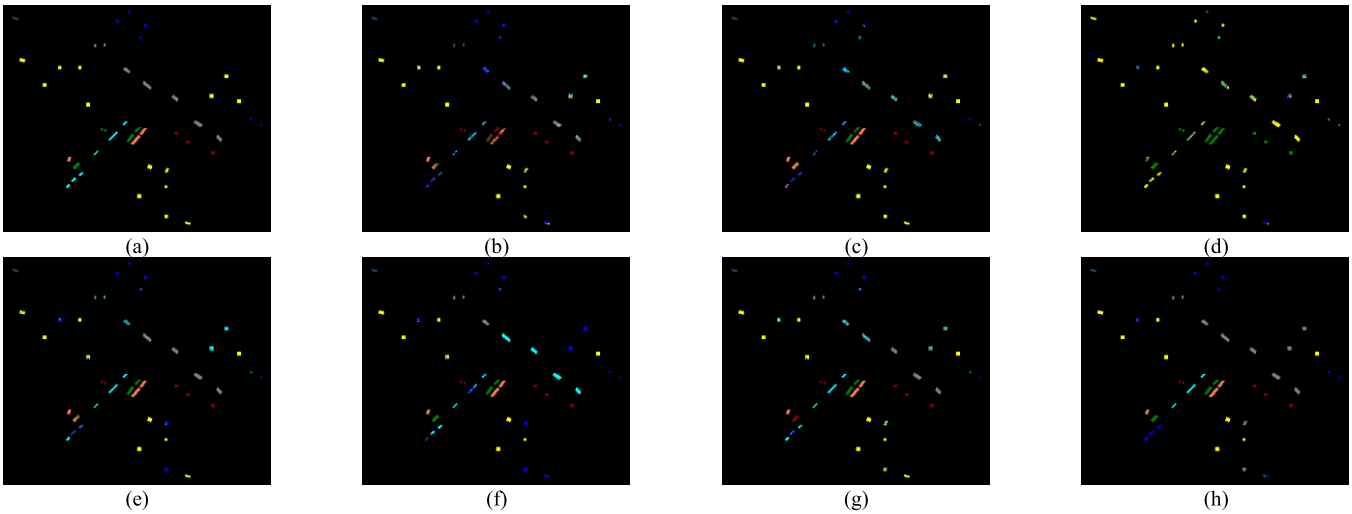



Fig. 11. Predicted labels of the shadow part of the Houston dataset. (a) GT. (b) JDA. (c) UTML. (d) GFK. (e) MEDA. (f) GEDA. (g) SSWK-MEDA. (h) MSF-MML.

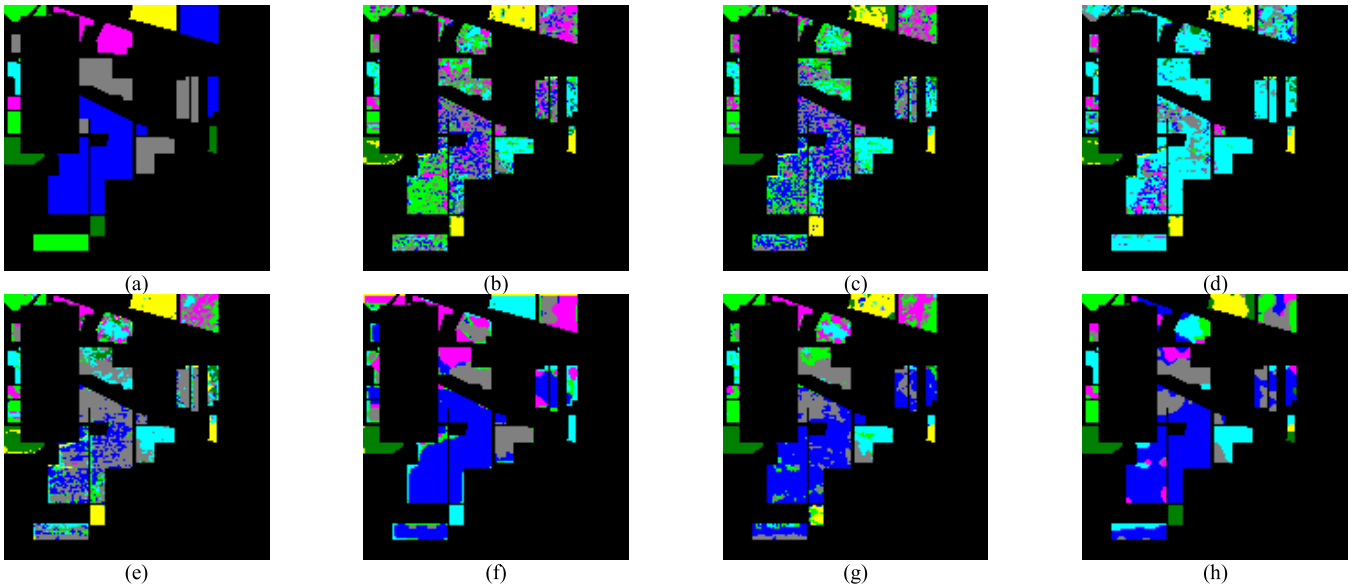



Fig. 12. Predicted labels of the big part of the Indian Pines dataset. (a) GT. (b) JDA. (c) UTML. (d) GFK. (e) MEDA. (f) GEDA. (g) SSWK-MEDA. (h) MSF-MML.

dataset, and the Worldview-2 dataset. To be fair, all methods use the same range of hyperparameters and choose the best score. It is easily observed that MSF-MML exhibits the best classification performance compared to the comparison methods. Compared to the distributed alignment-based methods GEDA, JDA, and UTML, first, MSF-MML uses manifold space special geometrical structures, which mitigate the problem of degenerated features. Then, MSF-MML shows that class-level alignment may be more beneficial for domain adaptation than GEDA and JDA. Compared to UTML, metric correction seems to be needed because suitable evaluation methods are lacking to weigh the importance of interclass and intraclass distances. Therefore, MSF-MML improves the OA by 0.86%–33.19% and kappa by 0.0107–0.3962 compared to GEDA, JDA, and UTML. Compared to the manifold subspace alignment-based method GFK, MSF-MML aligns class-level features, which makes up for the lack of domain-level feature alignment in GFK. Thus, MSF-MML improves the OA by 35.33%–60.72% and kappa by 0.3801–0.6886 compared with GFK. MEDA is an excellent method to effectively mitigate degenerated features and align domain distribution. However, it is not an algorithm designed for remote sensing, so it lacks algorithmic design for remote sensing data features, such as spatial features. MSF-MML both uses spatial features to participate in remote sensing image classification and also aligns domains through class-level features. Thus, MSF-MML improves the OA by 5.88%–34.71% and kappa by 0.0617–0.4194 compared with MEDA. SSWK-MEDA only uses a shallow spatial feature that fails to better utilize spatial characteristics. MSF-MML utilizes $L$-level spatial features to participate in the classification, which improves OA by 1.6%–21.37% and kappa by 0.0184–0.2543 compared with SSWK-MEDA.

*2) Qualitative Evaluation:* The classification results of these tasks are shown in Figs. 11–13. However, due to space constraints, we only show the "Bright→Shadow"

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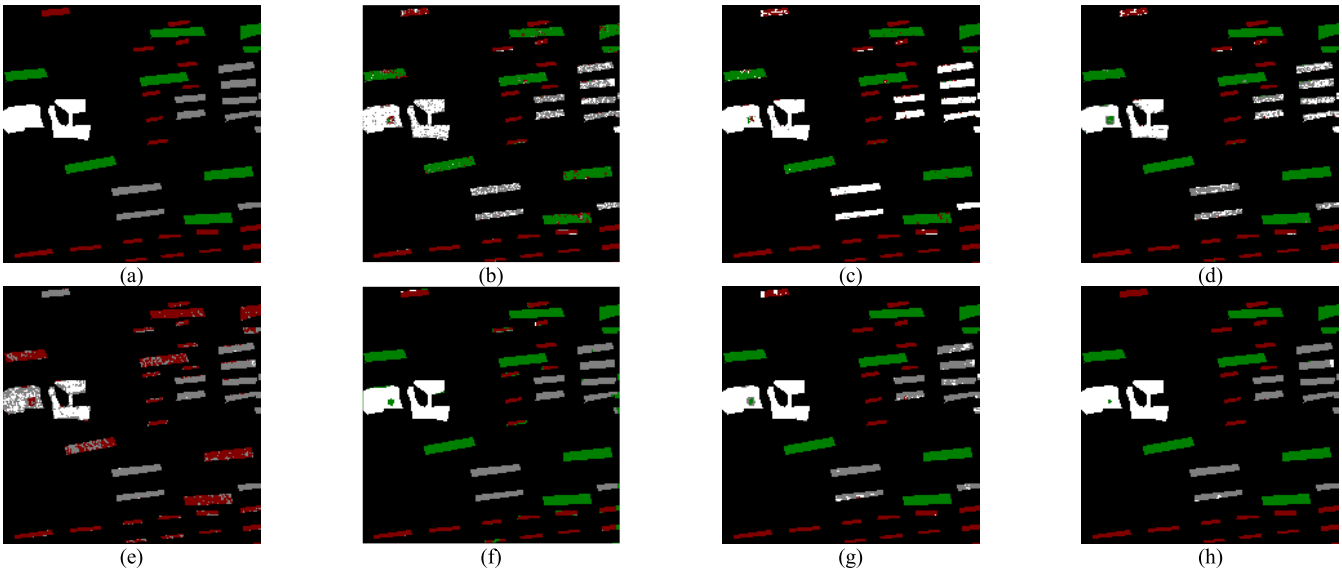



Fig. 13. Predicted labels of the 2011 part of the Worldview-2 dataset. (a) GT. (b) JDA. (c) UTML. (d) GFK. (e) MEDA. (f) GEDA. (g) SSWK-MEDA. (h) MSF-MML.

TABLE IV
TIME CONSUMPTION (SECONDS) OF DIFFERENT METHODS

| Dataset | Task | JDA | UTML | GFK | MEDA | GEDA | SSWK-MEDA | MSF-MML |
|---|---|---|---|---|---|---|---|---|
| Houston | Bright->Shadow | 269.6049 | 258.0562 | 0.5204 | 89.2266 | 45.7784 | 98.7341 | 114.6909 |
| | Shadow->Bright | 262.9120 | 270.9991 | 0.2159 | 88.0508 | 65.4944 | 92.6890 | 119.0140 |
| Indian Pines | Big->Small | 73.7504 | 79.1592 | 0.1654 | 24.4549 | 18.0331 | 25.3874 | 41.6453 |
| | Small->Big | 74.8249 | 77.5469 | 0.1874 | 24.9263 | 47.6019 | 26.4367 | 40.2919 |
| Worldview-2 | 2011->2012 | 292.3906 | 271.7154 | 0.2160 | 94.6302 | 29.9291 | 97.0248 | 116.0573 |
| | 2012->2011 | 291.5792 | 300.8930 | 0.2641 | 98.0654 | 29.4343 | 98.4517 | 115.1946 |

task for the Houston dataset, the "Small→Big" task for the Indian Pines dataset, and the "2012→2011" task for the Worldview-2 dataset. We can clearly see that the predicted images of MSF-MML are closer to the GT than those of other methods. In the Houston dataset, all comparison methods show weak classification results, but MSF-MML has better classification performance for highway and residential compared to other methods. In the Indian Pines dataset, it is easy to see that MSF-MML has the lowest confusion in the soybeans-mintill of blue and grass-pasture of green compared to the other methods. In the Worldview-2 dataset, it is evident that MSF-MML has good classification results, with only a few misclassified white and gray buildings.

*3) Time Complexity:* We also empirically test the time complexity of MSF-MML and compare it with other methods under the environment of Intel Core i7-8700 CPU and 16-GB RAM in Table IV. MSF-MML has time costs but does not generate wide time gaps compared with the comparison methods. The training of the final classifier takes a lot of time, which is the main reason for the time consumption of MSF-MML. However, we think that the time consumption is worth it with a significant improvement in classification accuracy.

In all comparison methods, GEDA, JDA, and UTML, based on distributed alignment methods, can show good performance in handling both domain shift and class distances. However, we believe that they may have ignored the presence of degenerated features and chose to perform feature transformation based on the original feature space, resulting in poor classification performance. GFK, based on the SA method, whose strengths are shown in the lower time costs and the alleviation of degenerated features, but the lack of class distances being constrained, leads to low classification results. MEDA is a superior algorithm in terms of dealing with domain shift, class distances, and alleviating degenerated features, but it does not take advantage of the important character of spatial features in remote sensing data. SSWK-MEDA exploits the strengths of MEDA while overcoming its weaknesses by including spatial features in the method, thus demonstrating excellent classification performance. However, the use of spatial features is simple, which constrains it from obtaining better classification results. MSF-MML makes full use of the advantages of the mentioned methods while striving to overcome their shortcomings. Furthermore, the effectiveness of MSF-MML is well demonstrated in a series of tasks with improved OA and kappa and the robustness of MSF-MML can also be well demonstrated with excellent performance in all tasks.

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## V. Conclusion

In this article, we propose a novel method called manifold ML with MSFs for remote sensing image classification. The proposed MSF-MML method utilizes a custom filter window to smooth the noise and obtain more stable feature information, which is called MSFs. Manifold mapping is a good trick for alleviating degenerated features in the original space, which helps MSF-MML improve the classification accuracy of the classifier. MSF-MML also innovatively uses metric correction to effectively reduce the differences in probability distribution between source domain and target domain and improve the robustness of the classifier. Several experiments illustrate that MSF-MML has excellent classification capability. We also explore the role played by each part of MSF-MML through ablation experiments, which demonstrates that MSF-MML has the best robustness. The consumption of time cost of MSF-MML is also beneficial along with the improvement of classification accuracy. Future work will center on leveraging the relationship between pixels adapted adjustment of weights between domain and class.

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